Alex Villa

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Education

University of Florida Gainesville, Florida
PhD in Electrical & Computer Engineering *August 2026 – May 2030*
Advisor: Professor Xiaoyi Lu

University of California, Merced Merced, CA
Bachelor of Science in Computer Science & Engineering, Minor in Applied Mathematics *August 2022 – May 2026*

Coursework: Algorithms, Numerical Analysis, Computer Architecture, Circuit Theory, Applied Statistics, Machine Learning, Numerical Linear Algebra, Parallel Computing, Operating Systems, Stochastic Processes
Affiliations: SIGHPC & SIGAPP, SIAM - CSE & Supercomputing.

Research Interests

High-Performance Computing (HPC), Parallel & Distributed Systems, Scientific Computing, Machine Learning, Physics-Informed Neural Networks, Collective Communication(MPI,NCCL), HPC Communication

Publications and Posters

Can Contrastive Learning Improve Class-Imbalanced Diffusion Model? [Submitted@ECCV]
Fang Chen, Alex Villa, Gongbo Liang, Xiaoyi Lu, Meng Tang *arXiv Link*

FleCSI: Flexible Computational Science Infrastructure [Submitted@JOSS]
Benjamin Bergen, Nick Moss, Irina Demeshko, Davis Herring, ..., Alex Villa, et al. *GitHub Link*

Accelerating Physics-Informed Neural Networks with Modern GPU Architectures Poster
Alex Villa, Yuke Li, Xiaoyi Lu *SIAM CSE 2025*

Experience

Lawrence Livermore National Laboratory Livermore, CA
Visiting Computing Scholar | Mentor: Dr. Daniel Nichols *May 2026 – August 2026*

- Implementing and optimizing communication-efficient collective operations via statistical modeling.

Los Alamos National Laboratory Los Alamos, NM
Software Systems Intern | Weapons Research Services - Weapons Mission Technology *May 2025 – Present*

- Partnered with customers to triage and resolve performance, compatibility, and configuration issues migrating HPC workloads to cloud platforms.
- Built a standardized internal framework for software lifecycle automation, cutting manual intervention by **87%** and downtime by **50%**
- Developing APIs and workflows to automate Granta MI user and property management across DOE NNSA.

Google-CAHSI Research Fellow Merced, CA
Undergraduate Research Assistant | Supervisor: Professor Meng Tang *January 2025 – August 2025*

- **Implemented CCUA in class-conditional diffusion models,** improving tail-class FID/Recall on ImageNet-LT (bias mitigation for long-tailed data)
- Refactored to distributed (data/model) parallelism(MPI/NCCL) and optimize compute/IO, **reducing training time by 66%**.
- **Introduce Unsupervised Contrastive Loss (UCL)**, employing unsupervised contrastive learning loss with negative samples only, enhancing within-class diversity.
- **Co-authored ICLR 2026 submission** on bias in image generation; contributed toward experimentation and analysis of the paper (under review).

Los Alamos National Laboratory Los Alamos, NM
Parallel Computing Research Intern | X-Computational Physics *June 2024 – August 2024*

- **Mentors:** Richard Berger, Davis Herring
- Led the design and implementation of a new feature introducing memory pointers as compute tasks within LANL's multi-physics framework, FleCSI.

- Refactored FleCSI's memory utilization strategy using the Legion programming model and C++17 to safely support new and ambiguous data types at both compile time and runtime.
- Optimized FleCSI with distributed-memory task execution across CPU and GPU architectures, ensuring safety, scalability, and high throughput.

Parallel and Distributed Systems Laboratory Merced, CA
Undergraduate Research Assistant | Supervisor: Professor Xiaoyi Lu *October 2023 – January 2026*

- Profiled and optimized large-scale training pipelines with *Horovod, MPI/NCCL*; instrumenting communication/computation overlap and I/O, reducing step time by **23.3%** across **>20** GPUs.
- Optimized irregular tensor ops (scatter/gather) via fused kernels, achieving **12**% higher throughput on A100/H100.
- Designed tailored collectives for irregular neural-operator workloads, improving load balance and model scaling.
- Create an in-depth research presentation highlighting optimization, analysis and future work in MPI, GPU Utilization & distributed memory usage for deep learning and molecular dynamic proxy applications(e.g. CabanaMD & miniMD).
- Researching and developing novel parallel approaches for Physics-Informed machine learning, enabling large-scale heterogeneous computing capabilities via CUDA and other programming models.

Cyberinfrastructure and Research Technology Merced, CA
Information Technology Consultant *October 2022 – May 2026*

- Provided skilled technical support for UC Merced's Pinnacle cluster, hosting office hours and introductory classes.
- Design and maintain documentation for UC Merced's Official HPC Cluster documentation.
- Core Developer & Instructor — UC Merced HPC Training: **Introduction to HPC & Intermediate HPC**

Technical Skills

Languages: C++, C#, Python, Matlab, Bash, CUDA(C++ & Python)
Frameworks: Horovod, MPI, NCCL, ROCm, Intel MKL, LLVM, Clang, CuBLAS, CuDNN, Legion, GASNet, Kokkos
Tools: Git, Containerization, AWS, SLURM, Google Firebase, HuggingFace, High Performance Computing, Distributed GPU Optimization Techniques, Cloud Computing, Transformers, Image Classification

Awards & Recognitions

National GEM Consortium — GEM Fellowship (PhD, full tuition & stipend, up to 5 years)
Los Alamos National Laboratory — Parallel Computing Fellowship ($10,000)
Google & CAHSI — Research Fellowship (2024–2025)
SC Conference — Student Volunteer Travel Grant (SC'25) ($1,500)
SIAM — Travel Award ($1,000)
SC Conference — IndySCC Travel Grant (SC'24) ($1,500)

Talks & Presentations

"Accelerating Physics-informed Neural Networks with Modern GPU Architectures" — SIAM CSE Poster Presentation, March 2025

"Horovod: For Heterogeneous Distributed Machine Learning Training" — Lightning Talk, SC24 Conference, November 2024

"FleCSI: A Framework for Task-Based Parallelism" — UC Merced Applied Mathematics: Data Science and Scientific Computing Seminar, October 2024

"Better Tasks for Task-Based Parallelism" — Los Alamos National Laboratory, August 2024

Leadership

IndySCC Team Lead at SC'24 and SC'25 University of California, Merced
Merced, CA *June 2024 - Present*

- Optimized HPC workloads by integrating AMD ROCm, HIP, and μProf to identify and resolve resource under-utilization.
- Accelerated High Performance Linpack(HPL) performance with MPI 5.0 collectives and topology-aware communication, reducing inter-node latency
- Increased molecular dynamics simulation throughput by 15% using GPU Residency and load balancing.